

Writing a Scientific Paper: Structure

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What Contains a Scientific Paper?

- A paper contains: CLAIM + ARGUMENTATION
- In other words: HYPOTHESIS + SUPPORT

It must convince that it is a CONTRIBUTION to the field:

- **relevance** (it addresses an important research problem)
- **originality** (it demonstrates new ideas useful for solving or understanding it)
- **validity** (the research work is convincing, well done)

IMRad (to be done)

Cambiar las transpas para adaptarlas al
<https://en.wikipedia.org/wiki/IMRAD>

Organization of a Scientific Article

Standard Structure

Allows to do **selective reading**: find quickly what you are looking for

- Title
- Abstract (short summary)
- Introduction (long summary)
- Preliminaries (context)
- Contribution (hypothesis, claim)
- Contribution (demonstration, evidence, argumentation for the claim)
- Related Work (context)
- Conclusions and Future work (learned lessons)
- References

Standard Structure

- Break the standard structure if there is a good reason for it:

- Being original is not a good reason

Number of readers of a paper in a good conference (educated guess):

- Title and Authors (300)
- Abstract (100)
- Introduction (50)
- Preliminaries (10)
- Contribution, ideas (10)
- Contribution, results (20)
- Related Work (10-20)
- Conclusions and Future work (30)

- **Title:**

- Try to be **sexy, concise** and **informative**
 - *Limited Discrepancy AND/OR Search and Its Application to Optimization Tasks in Graphical Models.*
 - *Of Mousetraps and Men: A Cautionary Tale*
 - *Boosting Search with Variable Elimination*

- **Authors:**

- Use the same name on all your papers
 - better short and simple but catchy
- In CS the order (relatively) matters

Limited Discrepancy AND/OR Search and Its Application to Optimization Tasks in Graphical Models

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If your paper was a movie, the abstract would be the **official trailer**

The purpose of the abstract is to allow readers to judge whether or not the article is of relevance to them. It is a **snapshot** of the paper.

- Single paragraph of 100-200 words
- Summary of scope, achievement and conclusions
- Self contained
- Written for as broad audience as possible
- Should not contain acronyms, abbreviations, mathematics
- Specific
 - Bad: *space requirements can be significantly reduced*
 - Good: *space requirements can be reduced by a 60%*

MINIMAXSAT: An Efficient Weighted Max-SAT Solver

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Abstract

In this paper we introduce MINIMAXSAT, a new Max-SAT solver that is built on top of MINISAT+. It incorporates the best current SAT and Max-SAT techniques. It can handle hard clauses (clauses of mandatory satisfaction as in SAT), soft clauses (clauses whose falsification is penalized by a cost as in Max-SAT) as well as pseudo-boolean objective functions and constraints. Its main features are: learning and backjumping on hard clauses; resolution-based and substruction-based lower bounding; and lazy propagation with the two-watched literal scheme. Our empirical evaluation comparing a wide set of solving alternatives on a broad set of optimization benchmarks indicates that the performance of MINIMAXSAT is usually close to the best specialized alternative and, in some cases, even better.

On the Practical use of Variable Elimination in Constraint Optimization Problems: ‘Still-life’ as a Case Study

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Abstract

Variable elimination is a general technique for *constraint processing*. It is often discarded because of its high space complexity. However, it can be extremely useful when combined with other techniques. In this paper we study the applicability of variable elimination to the challenging problem of finding *still-lives*. We illustrate several alternatives: variable elimination as a stand-alone algorithm, interleaved with search, and as a source of good quality lower bounds. We show that these techniques are the best known option both theoretically and empirically. In our experiments we have been able to solve the $n = 20$ instance, which is far beyond reach with alternative approaches.

Introduction

If your paper was a movie, the introduction would be the **plot** as in the Wikipedia

The Introduction is like an expanded version of the abstract. It should tell (and high-level explain) your achievements.

- The article's **topic**
- The **problem** being studied and why it is important.
- The **approach** to the solution
 - Consider giving an **example** if the main idea can be extracted out of it
 - Why this is **original**
 - Why it is **useful**
- **Scope** and limitations of the solution
- Relevant literature should be cited as needed (for credibility and context)
- Do not include complex definitions or mathematics

The Preliminaries are used to provide **background knowledge** (definitions, ideas from previous works) that is used in the paper.

- Always **cite the sources**, but you can change notation for your convenience.
- **Be concise** (assume the reader knows it, else he can go to the sources)
- Consider adding **examples** if they can be re-used later on (e.g. to illustrate your contributions)

fact

Sometimes I have fully understood a paper years later, when reading the preliminaries of a posterior paper mentioning it here.

BUCKET ELIMINATION

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Algorithm elim-mpe

Input: A belief network $BN = \{P_1, \dots, P_n\}$; an ordering of the variables, d ; observations e .

Output: The most probable assignment.

1. **Initialize:** Generate an ordered partition of the conditional probability matrices, $bucket_1, \dots, bucket_n$, where $bucket_i$ contains all matrices whose highest variable is X_i . Put each observed variable in its bucket. Let $S_1, \dots, S_j$ be the subset of variables in the processed bucket on which matrices (new or old) are defined.

2. **Backward:** For $p \leftarrow n$ downto 1, do

for all the matrices $h_1, h_2, \dots, h_j$ in $bucket_p$, do

- (bucket with observed variable) **if** $bucket_p$ contains $X_p = x_p$, assign $X_p = x_p$ to each h_i and put each in appropriate bucket.

- **else**, $U_p \leftarrow \bigcup_{j=1}^j S_j - \{X_p\}$. Generate functions $h_p = \max_{X_p} \prod_{j=1}^j h_i$ and $x_p^o = \argmax_{X_p} h_p$. Add h_p to bucket of largest-index variable in U_p .

3. **Forward:** Assign values in the ordering d using the recorded functions x^o in each bucket.

function BE(F, CC)

1. **if** $F = \emptyset$ **then** $\{CC$ contains a constant function $f\}$
2. $BestCost \leftarrow f$
3. **else**
4. $x_i \leftarrow \text{SelectVariable}(F)$;
5. $B_i \leftarrow \{f \in C | x_i \in \text{var}(f)\}$
6. $f_i \leftarrow \text{elim}_i(\sum_{f \in B_i} f)$;
7. $CC \leftarrow CC \cup \{f_i\} - B_i$;
8. $BE(F - \{x_i\}, CC)$;
9. $x_i \leftarrow$ best extension of the assignment
to $(x_1, \dots, x_{i-1})$ relative to $\sum_{f \in B_i} f$

The Related Work is used to compare the new results to similar results in the literature

- Research is usually an extension of or an alternative to previous work
- Research usually uses ideas, inspiration, from other works
- It is not always easy to relate different papers
- Puts yours and related work in common terminology
- Also helps readers to understand the work
- Points to standard references
- May appear after the Introduction (1), before the Conclusions (2) or diluted along the paper (3)
 - ① Helps to understand the technical contents of your contribution
 - ② Helps identify the originality of your work

- A small portion of the paper (i.e, 20-40% in a conference paper, 30-50% in a journal paper)
- The structure should be evident in the (sub-)section headings
- Narrative **flow** and **clear logical structure** is essential
- **It must include:**
 - Detailed claim and major original concepts (already presented informally in the Introduction)
 - Chain of reasoning that supports the claim
 - Details of contributions (central proofs, algorithms, definitions)
 - Summarize experimental setup
 - Important results (summarized in a table?)
 - Discussion of why results support the claim made by the paper

Usual Structures of the Contribution

- **Chain**

- When the motivation of one results comes from the previous one

- **By specificity**

- First give outline and then fill in details
- **Example:** Introduce an algorithm in terms of generic data structures and later on show the best choices (e.g. a particular type of hash table)

- **By example**

- First apply ideas to some typical case and then explain more formally
- **Example:** Introduce a search algorithm in the context of chess and then generalize

- **By complexity**

- First explain over a simple case and then generalize to the most general case
- **Example:** Introduce an algorithm in the Integers domain, and then generalize it to Rings (algebraic abstraction)

- Description and justification of the experiments (hardware, software, benchmarks,...)
- Quantitative Results (tables with data and relevant things that can be observed from the data)
- Interpretation of the results (what can be extracted from them) and critical confrontation with the Expected results.

- **Conclusion:** closing summary
 - Do not repeat the abstract
 - Can **look beyond** the current paper and be **speculative** about variations, extensions, consequences,...
- **Bibliography:** list of cited references and links
 - Most people do not read the papers they cite
 - Typos are propagated
 - Some people self-cite too much
- **Appendices:** holds bulky material that would interfere with the narrative flow, or material that most readers do not care about (unaggregated results, code, lengthy proofs)
 - Usually not necessary (consider giving a link instead)

Fact

An article is not a novel

- Do not keep secrets until the end
- *Spoilers* should be all over the place

Fact

An article is not a diary

- Do not get influenced on **how did you get there**
- Do not get influenced on **how much effort you put** on the different parts

- ① Think of **what do you want to say**: choose what to emphasize and what to discard
- ② A conference paper reports a sharp result (one idea)
- ③ A journal paper may group several related ideas (possibly from conference papers)

- ① Write a **template**: work out a logical sequence of sections and subsections (writing the titles)
- ② **Sketch each section** in 20-200 words
- ③ Write a first draft of the **Introduction**
- ④ Write a first draft of the sections containing the **contributions**
- ⑤ **Iterate** over the different sections **editing, deleting** and **adding**
 - During the first iterations do not worry much about style.
 - Along iterations move from the **what to say**, to the **how to say** it
 - Expect to dislike what you wrote the day before
- ⑥ Leave the **Abstract** and the **Conclusion** for the end.